2026

Tanner Antons

Network Scanner

AGENDA

01 The Problem

02 The Data

03 EDA

04 Preprocessing

05 The Model

The Problem

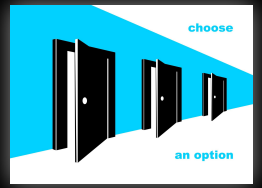
Given a set of network traffic logs with features describing connection metadata and
behavior, predict whether each connection instance represents a normal (benign)
event or a malicious (attack) event — and ideally identify the type of attack.

The Data

- CIC UNSW-NB15
- Canadian Institute for Cybersecurity
- 3540241 entries
- 84 columns
- Only 2 Days worth of Data

Attack Types

Analysis



Backdoor

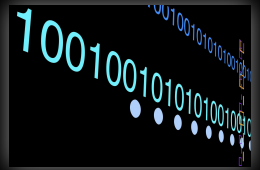
Denial of Service

Exploits

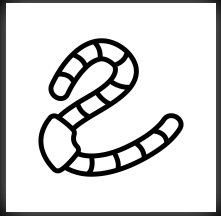
Fuzzers

Generic

Recon



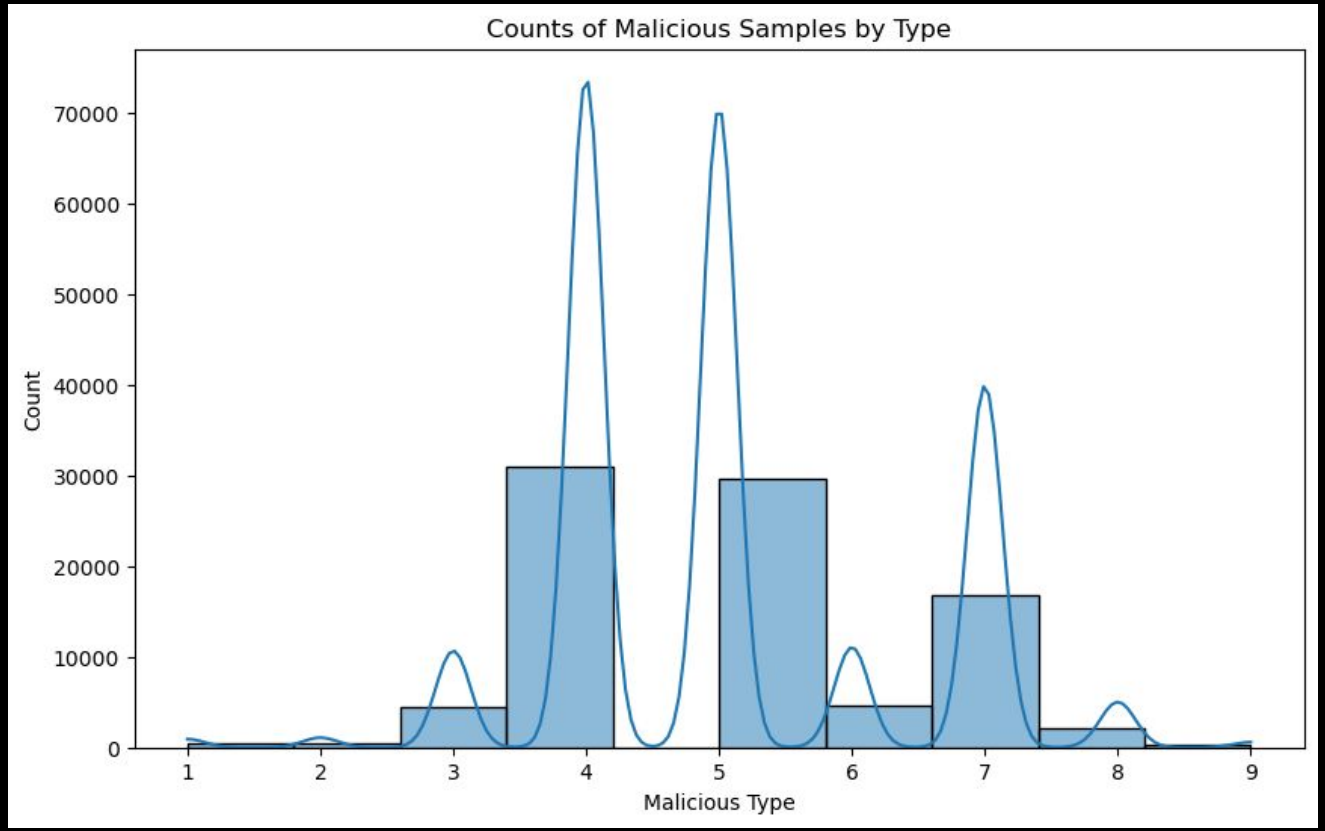
Shellcode



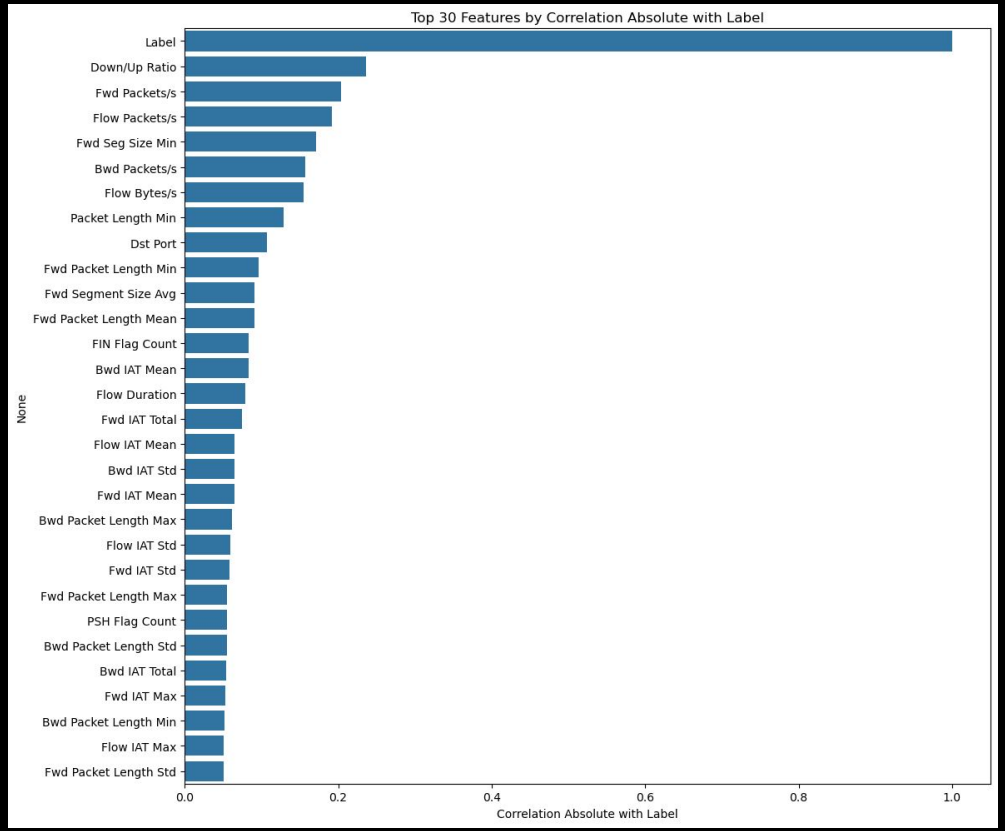
Worms

EDA or Exploratory Data Analysis

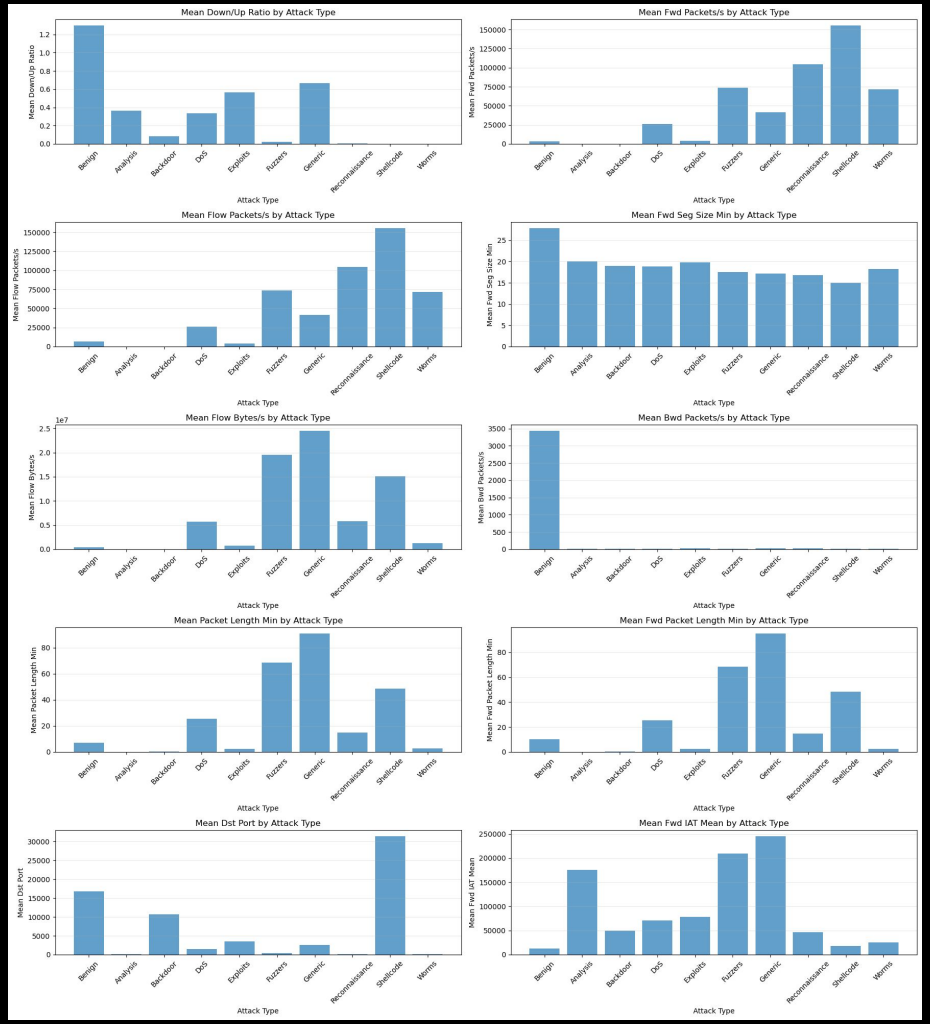
Benign	97,426
Analysis	11
Backdoor	12
DoS	128
Exploits	850
Generic	133
Recon	480
Shellcode	60
Worms	7
Type	# of Entries



EDA cont.


Top 30 Features by Correlation Absolute with Label
Label
Down/Up Ratio
Fwd Packets/s
Flow Packets/s
Fwd Seg Size Min
Bwd Packets/s
Flow Bytes/s
Packet Length Min
Dst Port
Fwd Packet Length Min
Fwd Segment Size Avg
Fwd Packet Length Mean
FIN Flag Count
Bwd IAT Mean
Flow Duration
Fwd IAT Total
Flow IAT Mean
Bwd IAT Std
Fwd IAT Mean
Bwd Packet Length Max
Flow IAT Std
Fwd IAT Std
Fwd Packet Length Max
PSH Flag Count
Bwd Packet Length Std
Bwd IAT Total
Fwd IAT Max
Bwd Packet Length Min
Flow IAT Max
Fwd Packet Length Std
None
0.0
0.2
0.4
0.6
0.8
1.0
Correlation Absolute with Label

EDA cont.


Mean Down/Up Ratio by Attack Type
Mean Down/Up Ratio
Mean Fwd Packets/s by Attack Type
Mean Fwd Packets/s
Mean Flow Packets/s by Attack Type
Mean Flow Packets/s
Mean Fwd Seg Size Min by Attack Type
Mean Fwd Seg Size Min
Mean Flow Bytes/s by Attack Type
1e7
Mean Flow Bytes/s
Mean Bwd Packets/s by Attack Type
Mean Bwd Packets/s
Mean Packet Length Min by Attack Type
Mean Packet Length Min
Mean Fwd Packet Length Min by Attack Type
Mean Fwd Packet Length Min
Mean Dst Port by Attack Type
Mean Dst Port
Mean Fwd IAT Mean by Attack Type
Mean Fwd IAT Mean
Attack Type
Benign
Analysis
Backdoor
DoS
Exploits
Fuzzers
Generic
Reconnaissance
Shellcode
Worms

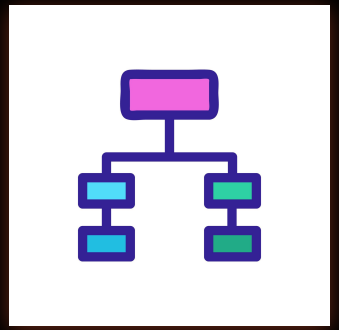
Preprocessing

Step 1

→ Take the 3 sample data sets made earlier and combine.

Step 2

→ Encode



Step 3

→ Split

All Data

Train

Test

Holdout

Step 4

→ Why we didn't scale?

The Model

The Five Models I started with.

The Macro F1 Scored for each Model.

Weighted F1 Score

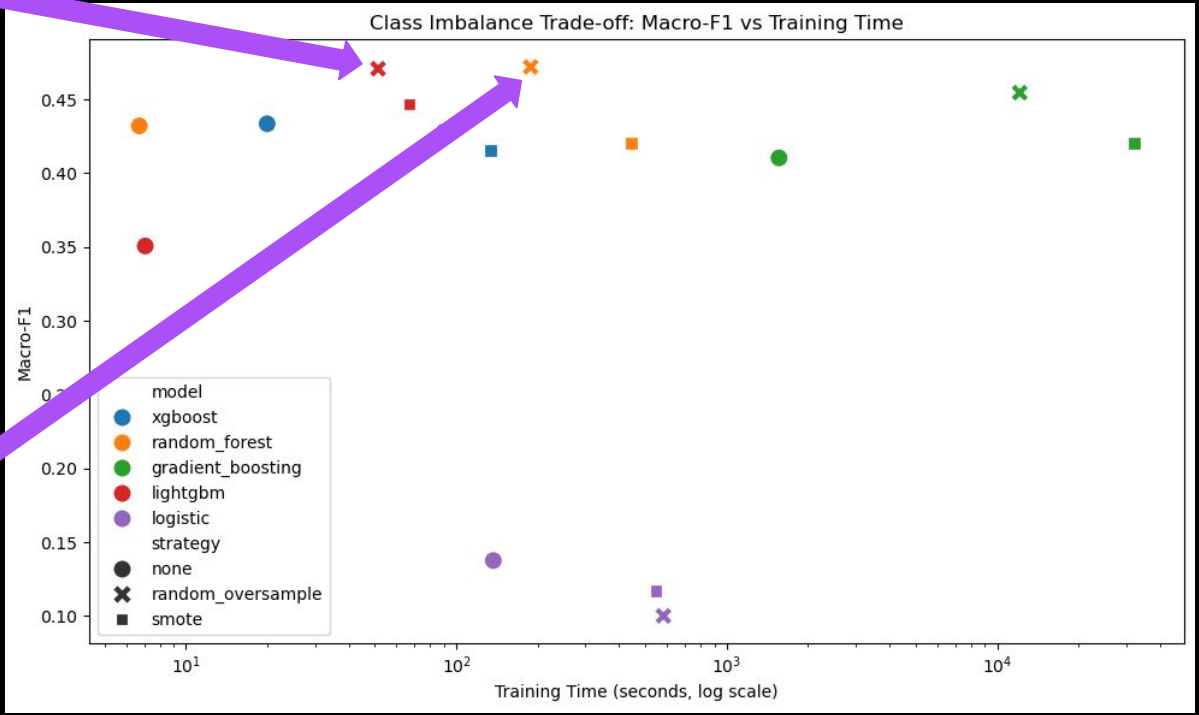
Time to create Model.

How fast it makes a prediction.

model	macro_f1	weighted_f1	train_seconds	predict_seconds	status
xgboost	0.4335	0.9845	21.45	0.724	ok
random_forest	0.4321	0.9824	6.86	0.626	ok
gradient_boosting	0.4104	0.9815	1190.97	2.553	ok
lightgbm	0.3507	0.9798	7.17	0.718	ok
logistic	0.1376	0.9548	83.19	0.021	ok

The Glaring Issue

LightGBM Random Oversample

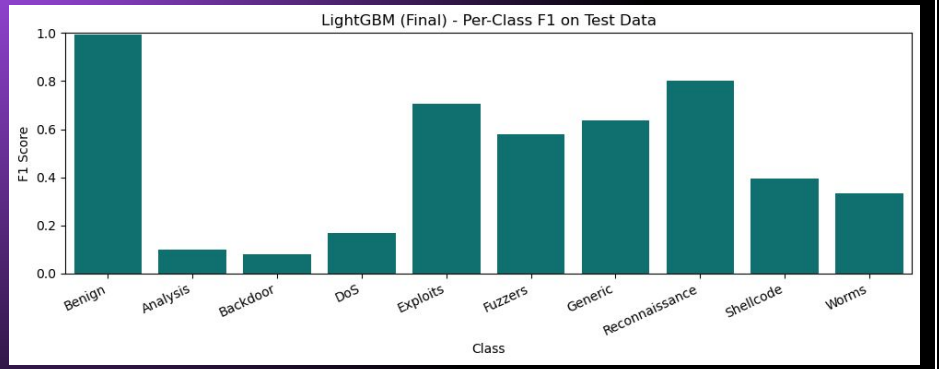


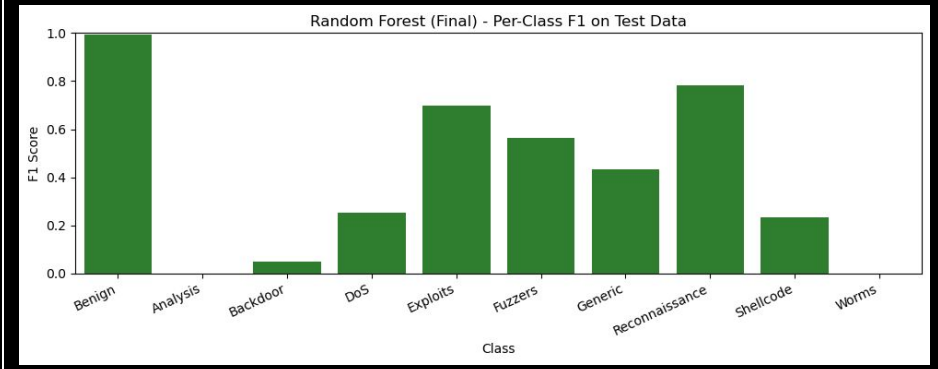
Random Forest with Random Oversampling

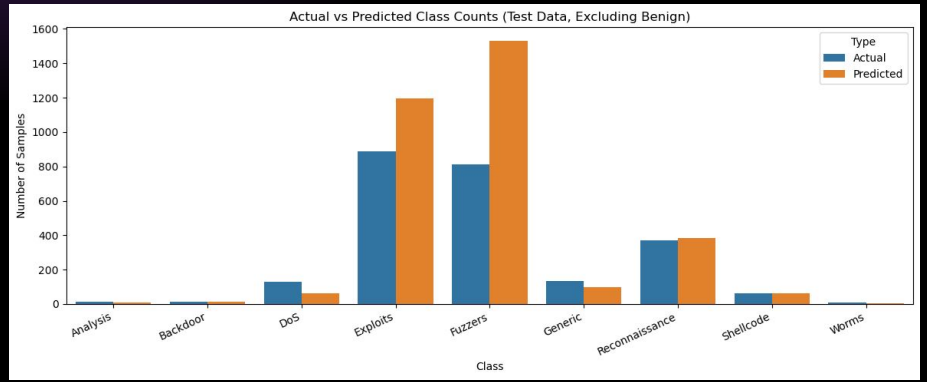
Tuning

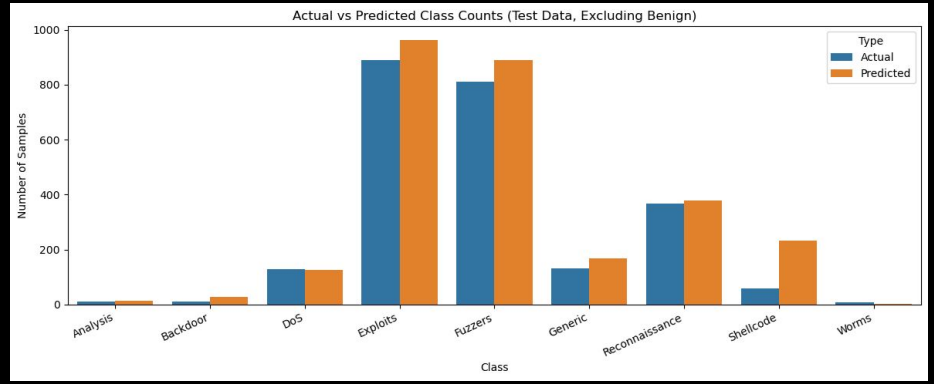
01	Speed	Untuned Side by Side comparison.
02	Imbalance	After testing Random Oversampling is the best solution.
03	Grid Search	Random Forest: Macro F1 = 95.84% LightGBM: Macro F1 = 96.21%
04	Closer Look	We need to see how the two models handle the smaller categories.

Side by Side Comparison





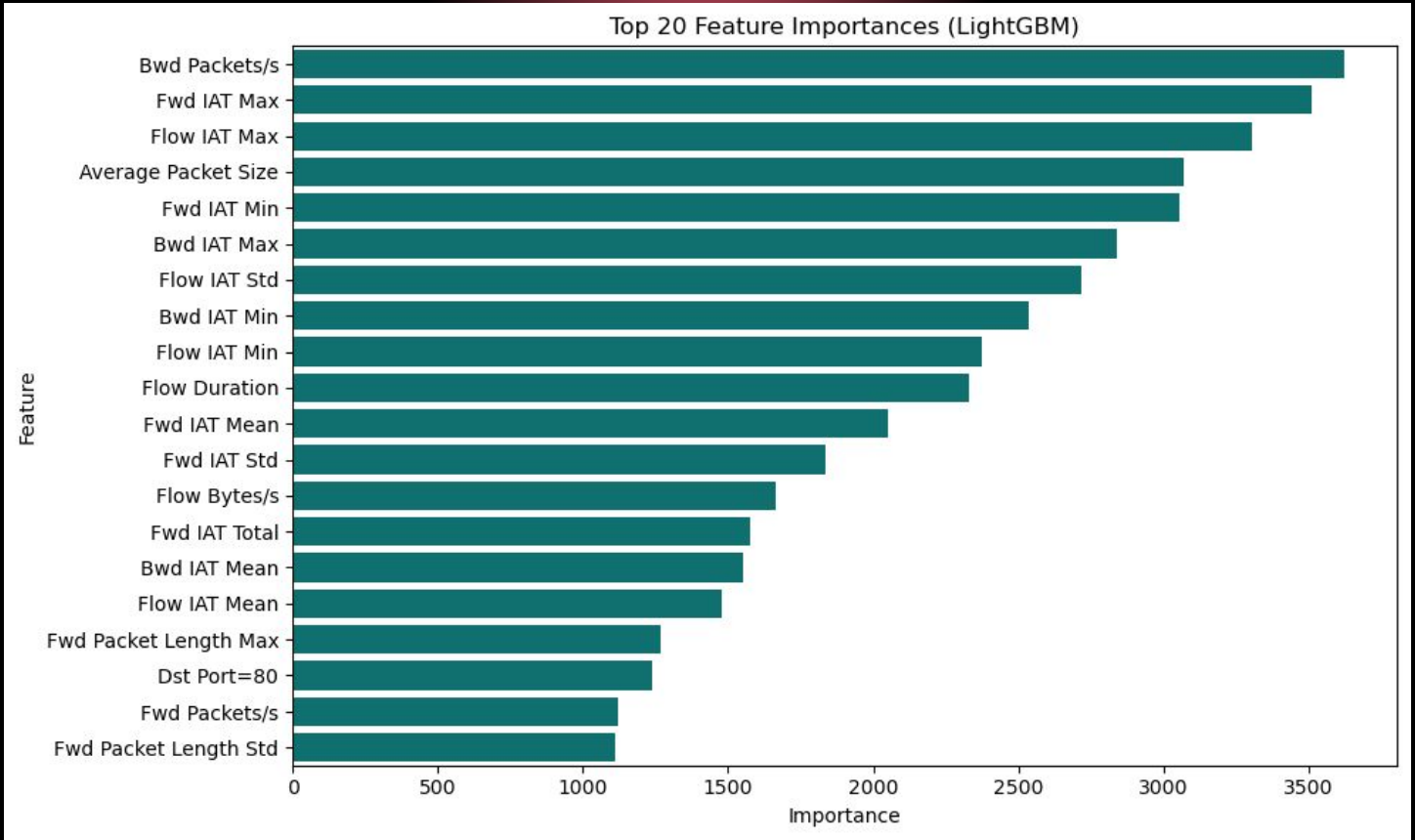




Why the

LightGBM Model?

Feature Importance and What's Next?



THANK YOU

Any questions?